

ARTS-Lab Graduate Student Handbook

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1 How to Use This Handbook

This handbook is designed to help students use their time in the lab efficiently and answer common questions about graduate study in the ARTS-Lab. It is not intended to be a highly structured set of rules, but rather a practical guide for supporting timely graduation.

2 Austin Downey's Perspective on Graduate School

Graduate school represents a pivotal phase in one's professional journey. It is a time for intensive learning, significant growth, and meaningful contributions. As a professor with dual roles of Advisor and Project Manager, I see graduate school through two critical lenses: 1) the pursuit of studies and 2) the financial responsibilities attached to them. The second point leads to thinking of graduate school as employment; albeit a job focused on your success not company's financial gain.

2.1 The Pursuit of Studies

Graduate school is an opportunity to develop deep technical skills in a chosen field while having the opportunity to explore complex problems and push the boundaries of knowledge. This demands dedication, hard work, and a genuine passion for research and learning. It is a period where you not only acquire advanced knowledge but also develop critical thinking, problem-solving skills, and the ability to work independently. The ultimate goal is to contribute new insights and innovations through scholarly work, culminating in published papers and presentations at conferences. This academic output is not just a measure of individual achievement but also a significant contribution to the broader scientific and academic community. As an advisor, my role is to help you through your academic journey. I provide intellectual support, help you navigate the complexities of your research, and offer feedback to ensure the quality and rigor of your work. I am here to help you develop your skills, identify opportunities, and overcome challenges in your studies. If your coursework is self-funded, the requirements for graduation are set by the Department of Mechanical Engineering Graduate Student Handbook.

2.2 Financial Responsibilities

Funding for graduate studies is very competitive and comes from sources through funded projects where I proposed a project and was awarded funds to carry out the research. When a sponsor provides financial support, it is not a mere act of charity. There are explicit expectations of return on investment. This return comes in the form of tangible outcomes, primarily publications. Broadly, these advancements hopefully lead to better lives for the citizenry, economic growth, and improved sustainability.

As a professor, I must emphasize the importance of understanding the differences between those who are self-funded and those who take financial support for their degrees. Accepting financial support carries a responsibility to deliver results. This means meeting project milestones, adhering to deadlines, and producing quality work that aligns with the sponsor's expectations. It is essential to view this relationship as a professional commitment, akin to a contractual agreement where both parties have obligations.

One crucial concept I stress is that graduation is not automatically granted when a student has been in the program for a certain amount of time or when funding runs out. Graduate school is not merely a matter of fulfilling course requirements as is done in undergrad. Rather it is about achieving specific academic and research goals. The culmination of graduate studies is marked by the successful completion of a body of work that meets the standards of the academic community

and the expectations of the sponsors. Simply put, taking someone's money obligates the recipient to produce a product. In the academic context, this product is papers.

As a Project Manager, my role extends to overseeing the financial and operational aspects of research projects. I ensure that the resources provided by sponsors are utilized effectively and that project timelines and milestones are met. This involves coordinating tasks, monitoring progress, and maintaining clear communication between all stakeholders. My responsibility is to ensure that the project delivers the expected outcomes on time and that all commitments to sponsors are fulfilled. We must honor the trust and investment made by our sponsors. It is crucial to fulfill these commitments fully, reflecting both the values and the responsibilities of our academic community.

2.3 Graduate School as Employment

I encourage all new group members to think of themselves as employees rather than students. This perspective carries a different set of obligations. As a student, you are essentially a customer. You pay for classes and, in return, receive academic credit and, eventually, a degree. Your primary obligation is to yourself, ensuring you get a good return on your investment by doing the work and achieving good grades. As an employee, you are a vendor. Others pay you and expect to receive a product, whether it be a paper, or other published product. This mindset shift can change how you approach your responsibilities.

To that end, the group has the following policy. Students must spend at least eight hours per day on campus Monday through Friday. This helps instill an employee-like mindset while enabling collaboration. Being present on campus during regular business hours allows for face-to-face interactions, fostering a professional environment and enhancing interactions between you and your colleagues. While remote work has become more common, this is an in-person lab.

It's crucial to challenge yourself, push your limits, and never settle for sub-standard work. By expecting more from yourself and others, you achieve higher standards and make the most of your graduate school experience.

3 Ph.D. Program Guidelines

3.1 Guidelines for Graduation

- Four published first-author papers in a peer-reviewed journal. At least three of these should be research-focused*.
- In certain cases, for competitive peer-reviewed conferences, a quality research-focused conference paper may be used in lieu of a journal paper.
- A good timeline to aim for is a conference paper in year 1, a journal paper in year 2, and subsequent journal papers in years 3 and 4.
- These papers should be published before the scheduling of the Ph.D. defense.
- A published conference paper is expected to set up the qualifier exam.
- A published journal paper is expected to set up the Ph.D. Proposal.
- To assist in making this timeline, it is advised that students do not frontload classes in their Program of Study. The last semester should be spend looking for jobs and answering review comments. The last semester is not for research and a course heavy load may be a good choice for this time.
- An expected version of the dissertation is paper-based, such that:
 - Chapter 1 is an introduction written for the dissertation.
 - Chapter 2 may be a review paper.
 - Chapters 3-5 may be journal papers or peer-reviewed conference papers.
 - Chapter 6 may be a conclusion written for the thesis.
- Any paper used in a dissertation will not be able to be used in another thesis or dissertation. For that reason, generally, only first-author papers will be allowed in the dissertation.

*Non-research-focused papers include hardware papers, reviews, topic overviews, position papers, and tutorials.

3.2 Ph.D. Qualifying Examination

The purpose of the Ph.D. Qualifying Examination is to ensure that students demonstrate the potential to successfully complete a Ph.D. It is an assessment of capability and preparedness for doctoral-level research (not a judgment of past accomplishments) and is therefore conducted early in the Ph.D. trajectory.

All students, without extenuating circumstances, will complete the Ph.D. Qualifier in the Dynamics & Controls Track. This track requires a written assignment provided up to two weeks before the ex-am and an oral interaction based on that assignment. The qualifier must be completed within 12 months of the first graduate course taken (24 months for APOGEE students).

The Qualifier Committee is composed of two or more faculty members. Dr. Downey will serve as one member, and the remaining members will generally be selected from the Department of Mechanical Engineering faculty who are positioned to support your research. These faculty members may or may not be members of your Ph.D. dissertation committee, as the qualifier and dissertation

committees are separate entities. The Qualifier Committee will administer the written and oral parts of the exam.

The written portion of the qualifier consists of a single integrated process: when the qualifier is scheduled, the student must send their published first-author conference paper to all committee members. Faculty may then choose to provide written questions. These questions may be related to the paper or may be general qualifier questions. These questions are typically sent about one week before the exam. Students must return written responses (if any are assigned), cc'ing Dr. Downey, no later than 48 hours before the oral portion of the qualifier. Failure to return written responses 48 hours before the exam will result in automatic cancellation of the qualifier.

The oral portion of the qualifier consists of a presentation, typically 20–25 minutes, followed by an extended question-and-answer session. In the oral portion of the qualifier, students are expected to demonstrate mastery of their research topic, a solid foundation in dynamics and control principles (or their field), and an ability to justify their methodological choices and conclusions. The presentation should address the student's broader research topic and not just the conference paper submitted for the written portion. The presentation should include the following list of sections. Ideally, in this order and with a clear section break for each section (It is ok if some sections only have one slide):

- **Student contributions:** In the opening slides, students should provide a complete list of their publications, categorized into journals, peer-reviewed conferences, conferences, abstract-only publications, and posters. Students must also include a clear statement of their individual contributions to the field, recognizing that contributions at the qualifier stage may be limited in scope. This discussion should emphasize the student's intellectual role and technical ownership of multi-author papers to articulate to the committee your contributions (rather than team member contributions). Contribution statements may be compiled and merged from the itemized list and the end of the introduction in each section of the student's papers.
- **Topic background:** An overview of the research area as sufficient to motivate the problem. This sets and establishes the necessary theoretical and practical foundations.
- **Literature review and related work:** A focused review and discussion of relevant prior work in the field, including a small set of key peer-reviewed journal papers closely related to the student's research efforts. A short review of three to five journal papers would be ideal. Students should demonstrate an ability to summarize, compare, and critically assess existing approaches, rather than provide an exhaustive survey.
- **Gap analysis and research challenges:** Identification of open challenges, limitations, or gaps in the existing literature that motivate ongoing and future research efforts. This section should demonstrate the student's understanding of what is missing in the field and why those gaps matter. This section is likely to be only one slide.
- **Current research direction:** This section is where the student presents their work, likely focusing on what was included in the paper submitted for the written portion and/or their broader research efforts. It should provide a high-level description of the student's developing research focus, including the types of questions being explored and the general methodological approaches being considered.

- **Future directions and feasibility:** A brief, forward-looking discussion of plausible next steps and longer-term research directions, intended to demonstrate that the student can reason about how a Ph.D.-level research program could be developed. This section should not include formal objectives, tasks, or timelines.

Students should send their presentation slides to Dr. Downey one week before the exam for re-view. Students are reminded that the qualifier is not a dissertation proposal: the goal is to demonstrate an ability to digest and understand complex research problems. Students are not expected to present complete solutions, but rather to show that they are capable of identifying meaningful problems and reasoning about how Ph.D.-level research tools and methodologies could be applied to address them.

Students are responsible for contacting their committee members, scheduling the exam time, and reserving a room. This practice is intentional: students should feel comfortable approaching committee members for support, updates, and research guidance, as committee interaction is fundamental to healthy research progress.

The Ph.D. Qualifier will result in one of three outcomes: pass, conditional pass, or fail. A pass indicates the student is prepared to proceed. A conditional pass requires the student to address specific concerns identified by the committee, which may include one-on-one follow-up meetings or written responses; the presentation will not be repeated. The committee will then determine an outcome of pass or fail, with all conditions resolved within two weeks. A fail indicates that the qualifier must be repeated in its entirety.

3.3 Ph.D. Comprehensive Examination

The comprehensive examination (also referred to as the proposal) evaluates a student's readiness to conduct independent dissertation research and judges their potential to complete a Ph.D., not merely their past accomplishments. Before scheduling the exam, the student must have at least one published or accepted first-author, research-focused journal paper that helps frame and motivate the dissertation proposal. The comprehensive examination must be completed within 24 months of the first graduate course taken (36 months for APOGEE students).

The comprehensive exam requires both a written proposal and an oral defense. The written proposal should be approximately 15 pages in length and generally follow NSF proposal format requirements (see the NSF Proposal & Award Policies & Procedures Guide (PAPPG) guidance for structure and expectations). The written document must include: a clear background and motivation, three research objectives, and three tasks (each task mapped to one objective). It should also contain a timeline for the proposed work and a summary of prior work as needed to justify feasibility. The proposal will need to be shared with Dr. Downey through a Dropbox folder for review. A template for the Ph.D. proposal can be found here <https://github.com/ARTS-Laboratory/ARTS-Lab-Ph.D.-Proposal-Template>. The oral proposal defense is approximately 30 minutes in length and should follow a structure consistent with the Ph.D. Qualifier, but at a more advanced and forward-looking level. The purpose of the comprehensive examination is to evaluate readiness for independent dissertation research, not to present a completed dissertation.

The presentation should begin with a concise summary of the student's scholarly record, including a categorized list of publications (journals, peer-reviewed conferences, conferences, abstracts, post-ers, etc.) and a clear articulation of the student's specific intellectual and technical contributions. As in the qualifier, contribution statements should reflect individual ownership and may be compiled from the introduction sections of published work, but they should be synthesized into a coherent statement of impact and research trajectory.

Following this opening, the presentation should include: (i) background and focused literature context sufficient to motivate the dissertation direction, (ii) a clear articulation of the overarching research objective and associated sub-objectives, and (iii) a structured description of the three proposed research tasks and their feasibility. The emphasis should be on technical clarity, logical alignment between objectives and tasks, and demonstration that the proposed work constitutes a coherent, Ph.D.-level research program. Unlike the qualifier, the comprehensive examination must present a well-defined and executable research plan. The committee will evaluate technical depth, feasibility, independence of thought, and the student's ability to defend methodological choices and anticipated contributions.

The written proposal must be sent to committee members one week before the scheduled exam date; failure to do so will result in automatic cancellation of the comprehensive exam. Additionally, students must send their presentation slides to Dr. Downey one week before the exam for review.

Students are responsible for contacting their committee members, scheduling the exam time, and reserving a room. This practice is intentional: students should feel comfortable approaching committee members for support, updates, and research guidance as committee interaction is fundamental to healthy research progress.

3.4 Ph.D. Final Defense

To schedule the final defense, students must satisfy all departmental and university requirements along with the group's publication expectations.

The final defense consists of a completed dissertation, a public presentation, and a closed-session question-and-answer portion with the committee. The public presentation is typically 40–45 minutes and should cover the overall research problem, gaps in the literature, the student's contributions, and the broader context of the work. It is crucial that students list all of their publications in the opening slides, clearly highlighting which publications are peer-reviewed, as consistent acceptance by peers provides strong evidence of research quality. Students must also list their contributions to the field (again, taken from their papers) and are encouraged to refer back to these contributions throughout the presentation. The primary purpose of the committee during the defense is to ensure that the student has made meaningful contributions to their scientific field.

All required publications must be accepted or published before scheduling. The dissertation must be sent to committee members one week before the scheduled defense date, and failure to do so will result in an automatic cancellation of the defense. Students must also send their presentation slides to Dr. Downey one week before the defense for review. Students are responsible for contacting their committee members, scheduling the defense time, and reserving a room, as this process encourages open communication and comfort in approaching the committee.

4 M.S. Program Guidelines

4.1 Guidelines for Graduation

- One published research-focused* first-author paper in a peer-reviewed journal.
- This paper should be published before the scheduling of the thesis defense.
- A good timeline to aim for is a conference abstract in semester 1, work on the conference paper in semester 2, expand to journal paper and submit in semester 3, and defend to semester 4.
- To assist in making this timeline, it is advised that students do not frontload classes in their Program of Study. The last semester should be spend looking for jobs and answering review comments. The last semester is not for research and a course heavy load may be a good choice for this time.
- An expected version of the thesis is paper-based, such that:
 - Chapter 1 is an introduction written for the thesis.
 - Chapter 2 may be a conference paper.
 - Chapter 3 may be a journal paper.
 - Chapter 4 is a conclusion written for the thesis.
- Any paper used in a dissertation will not be able to be used in another thesis or dissertation. For that reason, generally, only first-author papers will be allowed in the thesis.

4.2 M.S. Thesis Defense

M.S. students must publish or have accepted one first-author, research-focused paper prior to scheduling their thesis defense.

The M.S. thesis defense presentation should follow a structure similar to the Ph.D. examinations. Students must open their presentation by listing all of their publications, separated into journals, peer-reviewed conferences, non-peer-reviewed conferences, abstracts, posters, and other appropriate categories. Students must also clearly list their contributions to the field and are encouraged to refer back to these contributions throughout the presentation, as the primary role of the committee is to evaluate whether the student has contributed meaningfully to their area of study.

Students are responsible for contacting their committee members, scheduling the defense time, and reserving a room. The written thesis must be sent to all committee members one week before the scheduled defense; failure to do so will result in an automatic cancellation of the defense. Students must also send their presentation slides to Dr. Downey one week before the defense for review.

5 Frequently Asked Questions

5.1 Overview

The ARTS-Lab is an interdisciplinary research lab at the University of South Carolina that is comprised of individuals working towards a progression in their chosen field. The ARTS-Lab is the people that make up the lab, not space or equipment. As such, well-trained researchers are our product. The following guidelines are intended to answer questions those (graduate and experienced undergraduate students) involved in the ARTS-Lab may have.

1. Who do we work for?

- (a) The ARTS-Lab is funded almost exclusively by taxpayers. Therefore, we work to provide value to the taxpayers of both the United States and South Carolina.

2. How do we measure output?

- (a) Quality published papers. For students, papers are the currency of research.
- (b) Well-trained researchers. We are not a company focused on turning a profit.

3. How do I get my name on a paper?

- (a) A paper is a self-contained entity.
- (b) Names on papers are related to who wrote the text of the paper itself.
- (c) If you provided background work that was used in the development of the paper but did not contribute to the paper itself, you would not be an author.

4. How is the order of paper authorship determined?

- (a) Paper authorship order is in decreasing contribution to the text itself.
- (b) Typically, the position of students in the authorship order is prioritized.
- (c) Dual first-authorship notes may be placed on papers as required. However, the first listed name will be the person that contributed most to the text.
- (d) For papers that originate from our group, I (Dr. Downey) will be the contact author. This is done to reduce issues with author order.
- (e) If you graduate or leave the group, others may take over incomplete papers. As work progresses, authorship may need to be adjusted. However, authorship will still be determined by the order of decreasing contribution to the text itself.
- (f) Dr. Downey will make all final decisions on author order.

5. Who owns my/our work?

- (a) Generally, the taxpayers own your/our work. We provide them access to our work through publications and open-sourcing (as we can) our research products.
- (b) The ARTS-Lab may apply for patents on our work as a method to fund our work and future endeavors. As such, the University of South Carolina will own the exclusive right to our work for a set amount of time before it goes back to the public.

6. **I am leaving the lab (post-graduation, changing schools, starting a company) what can I take with me?**
 - (a) You are encouraged to take your work. You did it and I (Dr. Downey) am excited for you to use it.
 - (b) You may take published works and ideas. These are in the public domain just like any other published work.
 - (c) You may not take the work of others from shared databases or repositories. That is their work for them to use and share as they like.
7. **I don't feel comfortable sharing my work without knowing that I will be included in others' papers; I don't want to open-source my work; I feel my work is not being correctly rewarded?**
 - (a) The lab is not focused on keeping a "score" of who owes what to whom.
 - (b) We are focused on returning taxpayer investments to them through the training of researchers and the sharing of results. We are paid to deliver a product, we will deliver a product.
 - (c) The more we give, the more that will come back. Some ideas will feel like they are "lost" in the process, but that is part of life.
 - (d) Openness in research (within reason) is a core ethos of the ARTS-Lab. If this is an issue, please talk to me and we will find another group for you to work with.
8. **I have funding as an RA, why am I constantly being asked to apply for funding or work as a TA?**
 - (a) You will be asked to apply for funding opportunities as they come up to help spread the limited funds we have. However, papers are still the priority so additional loads are light.
 - (b) A good researcher is a self-funded researcher who employs others.
 - (c) Supplemental funding will not be in addition to any RA positions provided but will replace existing funds.
 - (d) We want to make sure there are funds available for others just starting.
 - (e) We all have side jobs to pay the bills.
9. **I am an undergraduate student who wants to switch labs; how should I handle this?**
 - (a) Please talk to me and we can find another lab for you and make suggestions. I am more than happy to help you find a lab with a better fit for you.
 - (b) My goal is to develop trained researchers, they don't have to be in the ARTS-Lab.
10. **What conferences will I get to go to/ what am I required to do?**
 - (a) Enjoy your time by being involved in the conference and building connections.
 - (b) You need to talk to others. Sit at random tables with others, ask participants about their research, ask friendly questions in presentations, and go to every presentation you

can.

- (c) Leave work till you get back; most of those emails can and should wait.
- (d) Stay off your phone.
- (e) Conference travel is a very expensive investment funded by taxpayer money.
- (f) Do not spend your time talking to others from the ARTS-Lab and limit your interactions with others from the University of South Carolina.
- (g) It is not a paid vacation for you to enjoy a city.

11. I am a senior graduate student, why do I keep getting paired with undergraduate students? I could do this work faster myself.

- (a) As you progress through your graduate studies, training others becomes an important part of the work you do.
- (b) Figuring out how to use the limited (and sometimes sporadic) skills of others is an invaluable skill.
- (c) An ability to help others progress is what separates great researchers from decent ones.

12. If we meet (post-graduation) at a conference, do I have to say hi?

- (a) No; but if you do not I will be very sad.

6 Software Stack

6.1 Overview

This guide is here to help you quickly and easily get started with writing, formatting, and managing your research documents, figures, and bibliographies. By following these simple guidelines, we can foster a collaborative environment, maintain consistency and professionalism, and reduce confusion or rework for everyone—especially those new to our team.

6.2 Programming

- Anaconda/Spyder/Python:
 - Highly preferred for plotting
 - We employ Anaconda, a free and open-source distribution of Python, to streamline package management and deployment.
 - For Python development, we utilize Spyder, an open-source integrated development environment (IDE) designed for engineers and scientists.
- MATLAB with Simscape: For modeling and simulating physical systems, we employ MATLAB alongside Simscape. Simscape allows for the creation of models within the Simulink environment, facilitating the simulation of electrical circuits, mechanical assemblies, and.
- LabVIEW: LabVIEW is our chosen graphical programming environment for data acquisition, instrument control, and industrial automation. Its intuitive visual interface enables seamless integration with hardware components, streamlining the development of complex measurement and control systems.

6.3 Document Templates

- Use our lab’s official repository to access PowerPoint templates, icons, and branding resources: <https://github.com/ARTS-Laboratory/ARTS-Lab-Branding>

6.4 Document Preparation

- LaTeX backends:
 - Windows: Use MiKTeX
 - Linux: Use TeX Live
- Editor: TeXstudio for writing and editing LaTeX documents.
- Reference Management:
 - Use JabRef for managing bibliography files (.bib).
 - Citation Key Pattern: [auth][year][shorttitle:titlecase]
 - Example: Downey2024ContinuousWaterQuality
- Use tiled “~” to create non breaking spaces between numbers and units, figures and numbers, equations and numbers, and before citations.

- Style considerations:
 - What text in equations is italic? <https://physics.nist.gov/cuu/pdf/typfaces.pdf>
 - To obtain non-italic units in matplotlib, consider using Unicode such as ($\mu\epsilon$) for $\mu\epsilon$.
 - Style for spaces around units: <https://www.nist.gov/pml/special-publication-811/nist-guide-si-chapter-7-rules-and-style-conventions-expressing-values#:~:text=7.2%20Space%20between%20numerical%20value,value%20and%20the%20unit%20symbol.>
- Sharing Files and Collaboration Workspace: We'll share a dedicated Dropbox folder for each publication. Dropbox makes it easy to rename and move files, helping everyone keep things organized and up to date. This also ensures that when you eventually leave the lab, you can retain your files and documents independently of university-owned systems. While Dropbox does require a subscription, it's currently the best solution we've found for maintaining a consistent workflow and enabling seamless collaboration.

6.5 Python Common Plotting Header

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-

import IPython as IP
IP.get_ipython().run_line_magic('reset', '-sf')
#%% import modules and set default fonts and colors

"""
Default plot formatting code for Austin Downey's series of open source notes/
books. This common header is used to set the fonts and format.

Header file last updated May 16, 2024
"""

import numpy as np
import scipy as sp
import matplotlib.pyplot as plt
import matplotlib as mpl

# set default fonts and plot colors
plt.rcParams.update({'text.usetex': True})
plt.rcParams.update({'image.cmap': 'viridis'})
plt.rcParams.update({'font.serif': ['Times New Roman', 'Times', 'DejaVu Serif',
'Bitstream Vera Serif', 'Computer Modern Roman', 'New Century Schoolbook',
'Century Schoolbook L', 'Utopia', 'ITC Bookman', 'Bookman',
```

```

    'Nimbus Roman No9 L', 'Palatino', 'Charter', 'serif']]
plt.rcParams.update({'font.family': 'serif'})
plt.rcParams.update({'font.size': 10})
plt.rcParams.update({'mathtext.rm': 'serif'})
# I don't think I need this next line as its set to 'stixsans' above.
plt.rcParams.update({'mathtext.fontset': 'custom'})
cc = plt.rcParams['axes.prop_cycle'].by_key()['color']
## End of plot formatting code

plt.close('all')

```


7 Style Guidelines

7.1 Figures and Illustrations

- **Software:** Use Inkscape for creating and editing vector graphics and figures.
- **Format:** For optimal results in online publications, save figures as PNG files with a resolution of at least 300 DPI to ensure clarity and detail. If the figure contains photographic elements, using JPEG format at high-quality settings is appropriate. Avoid formats like PDF and SVG for final publication, as they can lead to larger file sizes and potential rendering issues.
- **File size:** To maintain control over the quality of your published work and prevent excessive compression by publishers, it's advisable to keep your entire paper's file size below 5 MB, with an absolute maximum of 10 MB. This approach minimizes the risk of the publisher applying additional compression that could degrade image quality.
- **Capital letters in Figure Labels and Graphs:** In our publications, we primarily use lowercase letters for labels in figures and graphs to enhance readability and maintain a clean visual presentation. This approach aligns with common practices where capitalization is reserved for proper nouns and the beginning of sentences. By using lowercase for most labels, we can effectively highlight specific elements when necessary by employing capitalization, such as for software titles or major items within a figure. This selective use of capital letters draws attention to significant components without overwhelming the reader by using bold text in figures.
- **File Naming Conventions & the “making_figures” Folder:** For each paper, create a dedicated folder called `making_figures`. Within this folder, organize subfolders named exactly after each figure as it appears in your LaTeX document (e.g., `Figure1`, `Figure2`, etc.). Each of these subfolders should contain all data files, Inkscape files, and any other assets required to produce that specific figure. This approach not only enhances reproducibility, but also makes it much easier to share or update figures in the future.
- **Copy Figures into a “figures” Folder for LaTeX:** Once you finalize a figure in its `making_figures` subfolder, place a copy of the final image file into a separate figures folder that your LaTeX document will reference. This way, you always know exactly which version of the figure is included in your paper, and you can easily provide the correct files when submitting to a journal. Keeping the “making_figures” and “figures” folders separate also serves as a helpful sanity check, ensuring you don't accidentally reference an outdated figure from somewhere else.
- **No Spaces in Figure File Names:** When naming your files or folders, do not use spaces. This is because many servers and systems (especially those running Linux) can encounter problems interpreting file paths with spaces, which may cause unexpected errors. Use underscores or CamelCase instead (e.g., `Figure1_Draft` or `Figure1Draft`). This practice is not only good computing hygiene but also ensures broader compatibility and fewer issues during file sharing or publication.
- **Match Figure Labels to File Names:** To avoid confusion and ensure consistency, use the exact same name for your figure file and the corresponding `\label` in LaTeX. This way, when you

see the label in your document, you'll immediately know which file it references. This practice helps prevent errors and makes it much easier to locate and update figures.

- Guidelines for Crafting Figure Captions with Subfigures:
 - Structure and Punctuation:
 - * Begin with a concise description of the overall figure, followed by a colon.
 - * Use lowercase letters in parentheses—(a), (b), (c), etc.—to label each subfigure.
 - * Provide specific descriptions for each subfigure, separating them with semicolons.
 - * Use "and" before the final subfigure description.
 - * End the caption with a period.
 - Capitalization:
 - * Start the caption with a capital letter.
 - * Use lowercase for subfigure descriptions, except for proper nouns or specific terms requiring capitalization.
 - Example:
 - * Microstructural analysis of alloy samples, showing: (a) as-cast microstructure; (b) microstructure after annealing; and (c) microstructure after quenching.
 - * The caption starts with an overarching description of the figure's content: "Microstructural analysis of alloy samples, showing:".
 - * Each subfigure is labeled with a letter in parentheses and described succinctly.
 - * Semicolons separate the subfigure descriptions, with "and" preceding the final item to delineate each component.
 - * Additional Examples:
 - Stress-strain behavior of materials under different conditions, illustrating: (a) tensile test at room temperature; (b) tensile test at elevated temperature; (c) tensile test after heat treatment; and (d) tensile test after cryogenic treatment.
 - Fatigue crack propagation in aluminum alloys, presenting: (a) crack initiation site; (b) crack growth path; (c) final fracture surface; and (d) post-fracture microstructure.
 - Comparison of machining techniques, depicting: (a) conventional milling process; (b) laser-assisted milling; (c) ultrasonic vibration-assisted milling; and (d) hybrid machining process.
- Color usage in plots
 - Educational Resource: It is required that you watch the following video, which provides valuable insights into effective colormap selection: Understanding Colormaps in Matplotlib. Experts have thoroughly considered this topic, so it's beneficial to leverage their findings rather than create new solutions from scratch. https://www.youtube.com/watch?v=xAoljeRJ3lU&ab_channel=Enthought
 - Sequential Data: For representing data with a natural ordering, such as intensity or magnitude, use the 'Viridis' colormap. 'Viridis' is perceptually uniform, making it a

superior choice for accurately conveying data variations.

- Categorical Data: When visualizing categorical data, utilize the 'Tab10' colormap, which is the default qualitative colormap in Matplotlib. Employing 'Tab10' ensures a consistent and distinguishable color scheme across your figures.
- Standardized Implementation: To facilitate consistency and ease of use, a standard Python code header is provided at the end of this document. This script sets the default colormaps and handles related configurations, streamlining the plotting process.
- Font Sizes in Figure
 - Consistency with Document Text: Ensure that the font size and style in your figures match those used in your main document. Typically, this means using 10 or 11-point Times New Roman font to maintain a uniform appearance.
 - Explicit Figure Dimensions: When creating figures in Python or Inkscape, set the figure dimensions explicitly to the desired size. Import them into LaTeX using specific width settings, such as `width=6.5in`. Avoid using scaling commands like `\textwidth=\linewidth`, as they can lead to inconsistent font sizes and misaligned elements.
 - Avoid Scaling During Import: Do not scale figures up or down during the import process. At each stage—creation, export, and inclusion—consider your figure's size, resolution (DPI), and font settings. Always define these parameters explicitly to ensure clarity and consistency.
- Labeling:
 - Create all figure labels using Inkscape to ensure consistency.
 - Use white text boxes with black text for clarity.
 - Avoid adding black outlines to label boxes to maintain a clean appearance.
- Arrowheads:
 - Use triangular arrowheads to indicate flow in diagrams such as flowcharts or pipelines.
 - For labeling specific items, use arrows with rounded back ends to point to the item.
 - Ensure arrows are either white or black, depending on the figure's color scheme, to maintain visibility.

8 Guidelines for EMCH 764

8.1 Introduction

Students register for EMCH 764 as a graduate-level independent-study course focused on producing a publishable journal or conference paper. The final paper is the sole basis for the course grade. In general, it is recommended that students focus on publishing a conference paper with a hard deadline either during or shortly after the semester for which they take the EMCH 764 course.

8.2 Enrollment limits Per University Policy

- Master's students may enroll in up to two distinct EMCH 764 courses during their Master's program.
- PhD students may enroll in up to three distinct EMCH 764 courses during their PhD program.
- If a student takes two EMCH 764 courses as a master's student and later continues to the PhD program and elects to apply those Master's EMCH 764 courses to their PhD Program of Study as a block, the student may take up to five total EMCH 764 courses (i.e., the two from the master's count toward the PhD limit when approved as a block).

8.3 Guidance for Completing the G-ISC Form

- **Course Description:**
 - State that the purpose of the independent study is to produce a journal-ready or conference-ready research paper. A concise entry students may use: “The objective of this independent study is to develop and complete a research project on [pick topic] resulting in a publishable paper to be submitted to [pick conference or journal].”
- **List of Tasks , Anticipated Submission Dates and Weighted Contribution to an Earned Grade:**
 - Students should provide a simple 3–4 sentence description of the planned work for the semester. This should briefly state the research topic, the general activities they expect to complete (e.g., background review, methods development, analysis), and that the final product will be a complete paper. Students should note that the entire course grade is based solely on the submitted journal or conference paper. A concise entry may read: “This independent study will involve background research, development of methods, completion of analyses, and preparation of a full paper for submission. The final grade is based entirely on the completed paper.”
- **Grading Scale**
 - A: ≥ 95 , A-: ≥ 90 & < 95 , B: ≥ 85 & < 90 , B-: ≥ 80 & < 85 , C: ≥ 70 & < 80 , C-: ≥ 65 & < 70 , D: ≥ 60 & < 65 , D-: ≥ 55 & < 60 , F: < 55